

THERMAL ENGINEERING - I

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): $2 \times 10 = 20$

- ② i) Which of the following is the extensive property of a thermodynamic system?
 a) pressure ☒ b) volume c) temperature d) density
- ① ii) In thermodynamics, heat and work are -
 a) point functions ☒ b) path functions c) both point and path functions d) non of these
- iii) Which of the following represent an open system?
 a) an air-tight container ☒ b) Pressure cooker c) a beaker full of water d) thermos flask
- iv) The work done is minimum in the following gas process.
 a) isobaric b) isochoric c) isothermal d) adiabatic
- ① v) Which of the following law of thermodynamics is used to understand the concept of energy conservation?
 a) zeroth law ☒ b) first law c) second law d) none of these
- vi) An ideal gas is one which obeys the law -
 a) $pV=RT$ b) $p = \frac{vR}{T}$ c) $pV = \frac{R}{T}$ d) none of these
- vii) A perfect gas at 300 K is heated at constant pressure till its volume is double. The final temperature is -
 a) 150 K b) 450 K c) 600 K d) No change of Temperature
- viii) At critical point, the enthalpy of vaporisation is -
 a) dependent on temperature only b) maximum c) minimum ☒ d) zero
- ix) The dryness fraction (x) of fully wet steam is -
☒ a) $x = 0$ b) $x = 1$ c) $0 < x < 1$ d) $x > 1$
- x) If the enthalpy of saturated water and dry saturated steam at certain conditions are 1185 kJ/kg and 2790 kJ/kg respectively, then the enthalpy of wet steam of dryness fraction 0.5 is -
 a) 1185 kJ/kg b) 3975 kJ/kg c) 2580 kJ/kg d) 2790 kJ/kg
- ① xi) In a fire-tube boiler, the water is in -
 a) tube ☒ b) shell c) all of these d) non of these
- ① xii) The natural draught in boiler is produced by
 a) steam jet b) centrifugal fan ☒ c) chimney d) both (a) and (b)
- ① xiii) An economiser is installed in a boiler primarily to -
 a) superheat the steam b) reduce fuel consumption c) increase steam pressure d) all of these

- ① xiv) The amount of water evaporated 'from and at 100 °C' into dry saturated steam at standard atmospheric pressure is called-
- a) evaporative capacity b) generation factor c) boiler horse power d) equivalent evaporation
- ① xv) The safety device used to protect the boiler when the water level falls below the minimum level in the boiler is-
- a) fusible plug b) safety valve c) blow-off cock d) water level indicator

GROUP-B

- ② 2. a) State first law of thermodynamics and specify its applications.
- b) A nozzle is a device for increasing the velocity of a steadily flowing stream. At inlet to a certain nozzle, the fluid parameters are: Enthalpy is 2850 kJ/kg; velocity is 50 m/s; area is 0.1 m² and specific volume is 0.18 m³/kg. At the discharge end, the Enthalpy is 2650 kJ/kg and the specific volume is 0.49 m³/kg. The nozzle is horizontal and there is negligible heat loss from it. Find the velocity of fluid at exit from the nozzle, mass flow rate and the exit area of the nozzle. [3+5]
3. a) What is thermodynamic system? Classify it and briefly describe them with examples.
- b) What is perpetual motion machine of 2nd kind (PMM2)? Write the efficiency of PMM2. Why such a machine cannot be constructed in actual practice? [(1+3) + (2+1+1)]
- ① 4. a) State the differences between real gas and ideal gas. Under which condition does a real gas behave most like an ideal gas? State the condition at which steam behaves almost like a gas.
- b) Show that $C_p - C_v = R$ [(2+1+1) + 4]
5. a) Derive an expression for the work done during an isothermal process of an ideal gas.
- b) A quantity of gas has a volume of 0.14 m³, pressure 1.5 bar and temperature 100°C. If the gas is compressed at constant pressure, until its volume becomes 0.112 m³, determine (i) the temperature at the end of compression and (ii) work done in the process. [4 + (2+2)]
- ② 6. a) List out the advantages of superheated steam.
- b) Determine the quantity of heat required to convert 1 kg of water at 30 °C to steam at a pressure of 6 bar under the following conditions:
- when the steam is dry saturated;
 - when the steam is wet having a dryness fraction 0.85; and
 - when the steam is superheated to 250°C, Take the specific heat of water as 4.2 kJ/kg K and that of the superheated steam as 2.25 kJ/kg K. [2+6]
- ② 7. a) Explain the terms 'critical point' with respect to steam generation.
- b) Write short notes on 'Mollier diagram'.
- ② c) A boiler evaporates 3.6 kg of water per kg of coal into dry saturated steam at 10 bar. The feed water temperature is 32° C. Find the equivalent evaporation from and at 100° C. [2 + 3 + 3]

⑥ 8. a) Classify boilers based on (i) relative position of hot gas and water, (ii) method of circulation of water.
Give example of each.

b) Describe with a neat sketch the working principle of Babcock & Wilcox boiler. [3 + 5]

9. a) State the functions of safety valve and blow-off cock employed in boiler.

b) State the advantages of artificial draught system over natural draught system.

c) The following data pertain to a steam power plant:

Height of chimney = 30 m ;

Draught produced 16.5 mm of water gauge ;

Temperature of flue gas = 360°C ;

Temperature of boiler house = 28°C ;

Determine the quantity of air used per kg of fuel burnt in the boiler. [2 + 2 + 4]